HR Contract Generator Documentation

Overview of Approach

My approach is built on a simple reality: software should handle the repetitive data entry, but humans must retain control over sensitive legal documents. I designed the software to do the heavy technical lifting behind the scenes, so the user only interacts with a simple, familiar interface automating the busywork without removing human judgment where it matters most.

The objective was simple: allow the HR team to generate contracts accurately and quickly. To achieve this, the system had to be completely intuitive and require zero technical training

Because of this, I made extreme user-friendliness my absolute priority. Here is how I approached it:

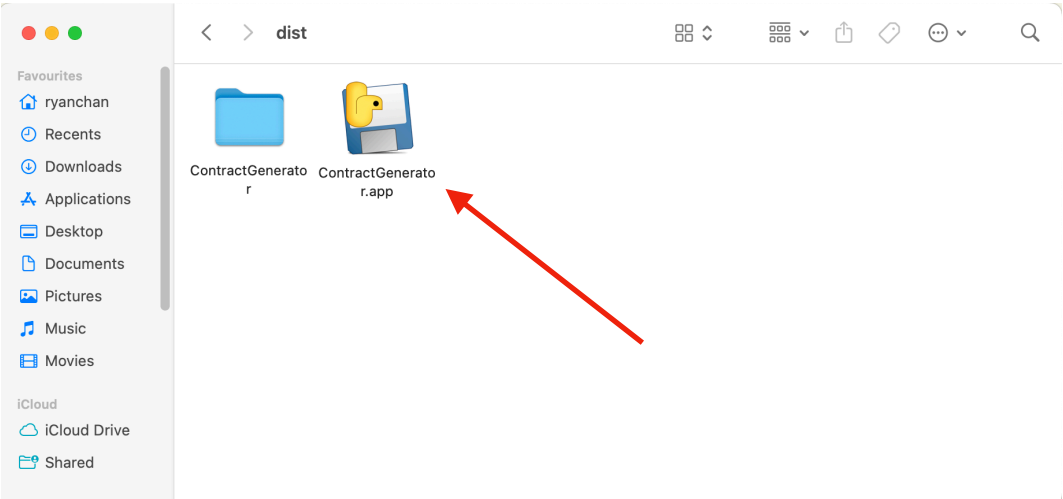
- **User-Centric Design:** I prioritized the actual needs of the HR department, recognizing that complex workflows fail if they aren't intuitive. I focused on exactly what the HR team needed. By keeping the interface straightforward and intuitive, I made the tool easy to learn and hard to make mistakes on.
- **Sustainable Architecture:** Instead of hardcoding a brittle solution, I built a flexible pipeline. I designed the system so it can adapt to future changes. Because the Word templates pull data directly from Excel, the HR team can update the contract formats themselves at any time, no developer help required.
- **Risk-Mitigated Efficiency:** I automated the repetitive work, like perfectly matching the ranks across Annex B and Annex C. But because these are sensitive legal documents, I made sure a human always reviews and approves the final contract before it is exported. This speeds up the process while keeping HR in full control. This balances operational speed with the strict quality control required for sensitive legal documents.

Tool Used

- **Python** was used as the main language because it is fast to build with, easy to maintain, and has strong document-processing libraries.
- **Tkinter** was used for the desktop interface because it is lightweight, practical, and suitable for a simple internal tool with minimal setup.
- **openpyxl** was used to read Excel files, since the workflow depends on Annex B and Annex C data being extracted directly from .xlsx files.
- **python-docx** was used to read, edit, and generate Word documents, since the output contracts are based on a .docx template.
- **PyInstaller** was used for packaging, so the application can be shipped as a standalone executable or app bundle without requiring the end user to install Python.
- **Microsoft Word** is used for PDF export. This allows the app to support PDF generation when those tools are available, while still falling back to .docx export if they are not.
- A local desktop application was chosen instead of a web app so the tool would be easier for HR to run internally, easier to distribute, and simpler to operate without server setup.

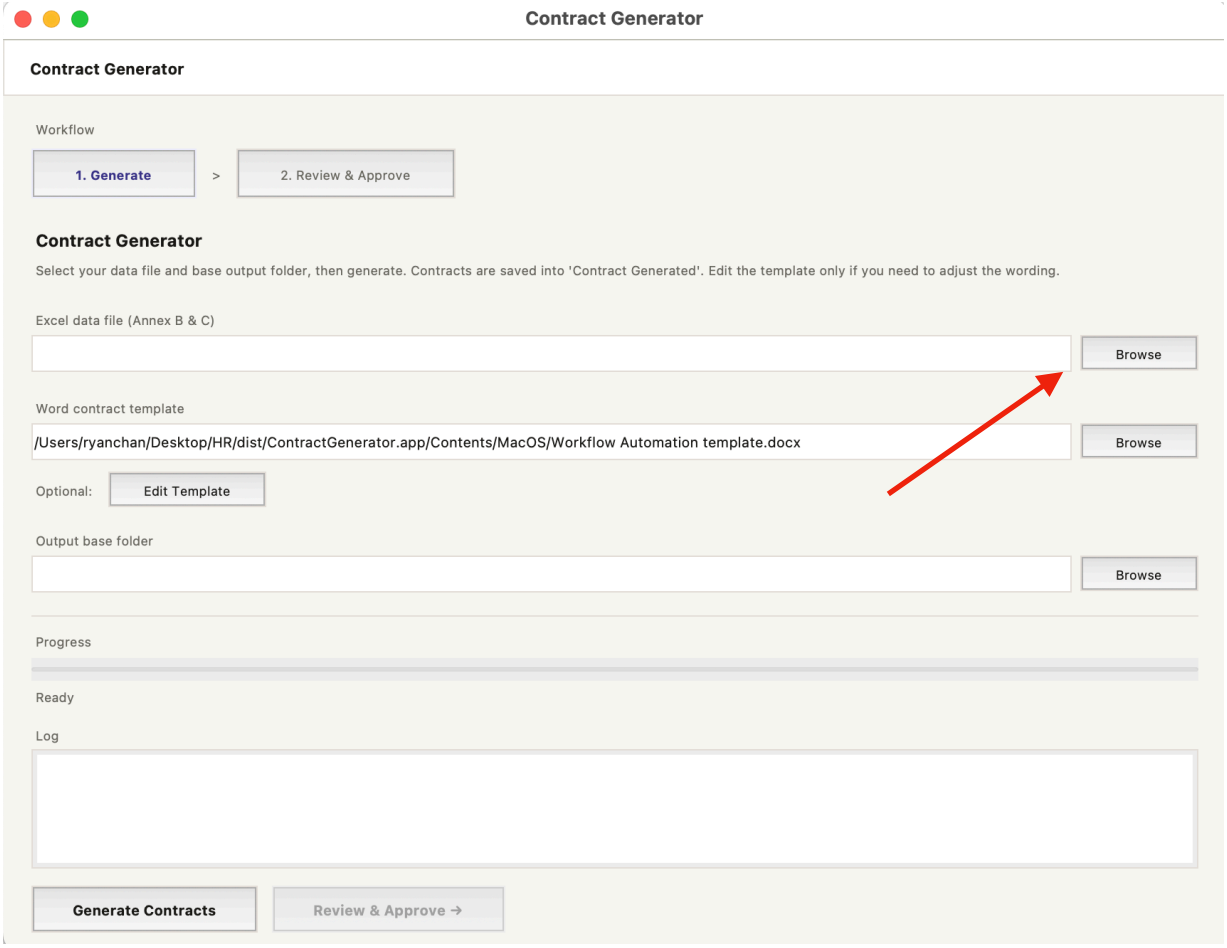
Step By Step Usage Guide (Read Setup Guide before proceeding)

First step (launching the application):



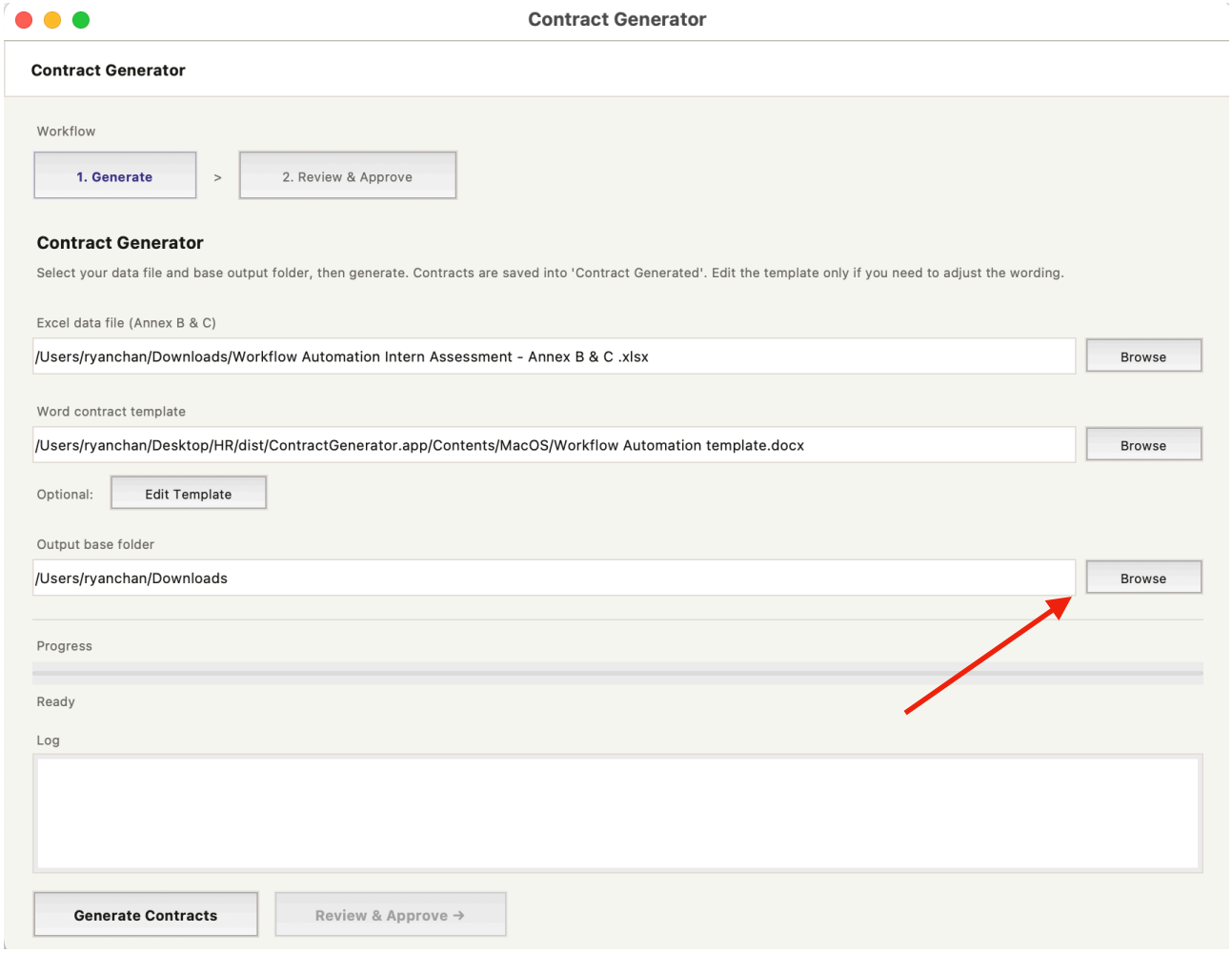
Click ContractGenerator.app for MAC
Click ContractGenerator.exe for Windows

Second Step (uploading candidate data):



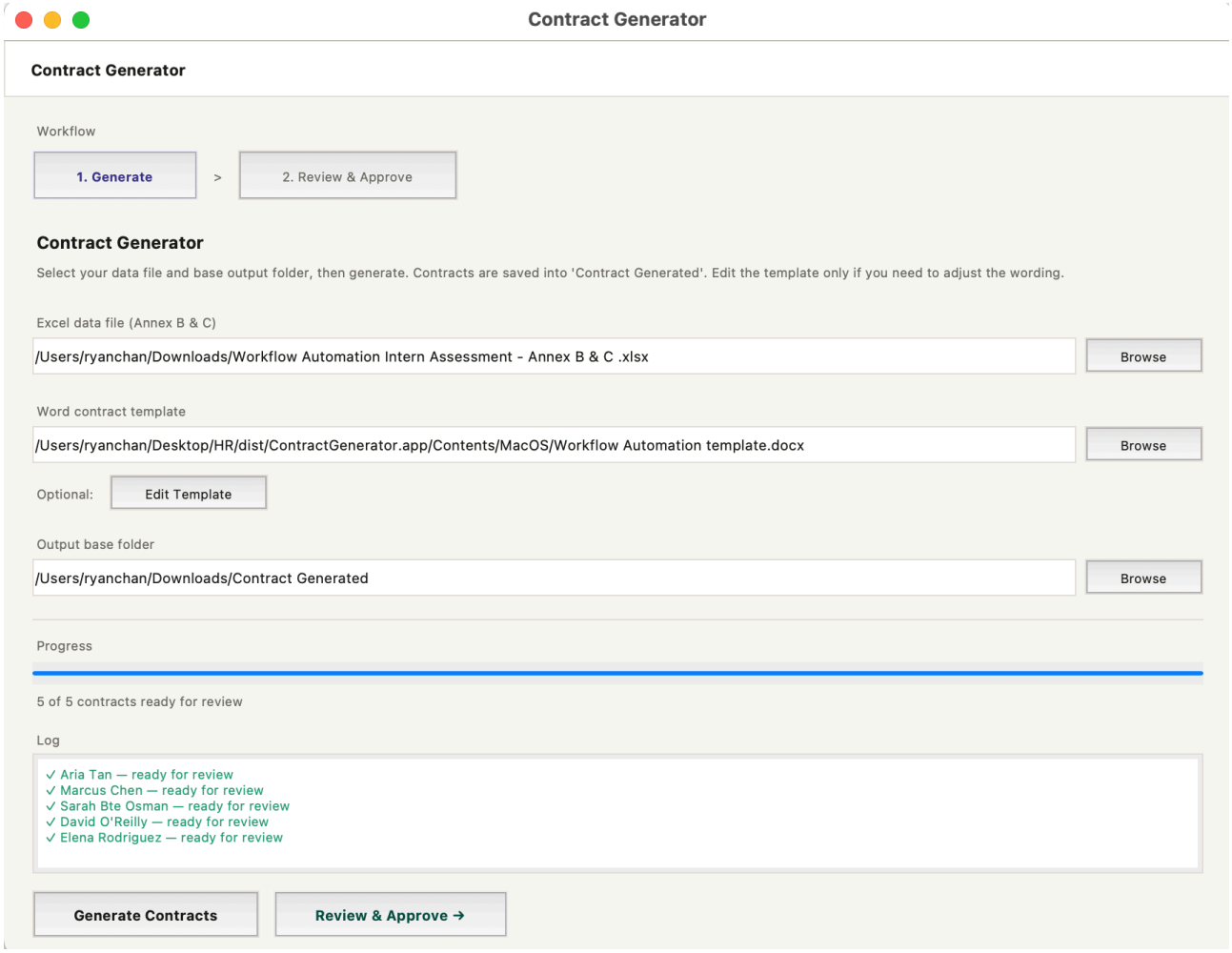
Click Browse to add Annex B & C candidate xlsx files

Third Step (selecting destination folder):



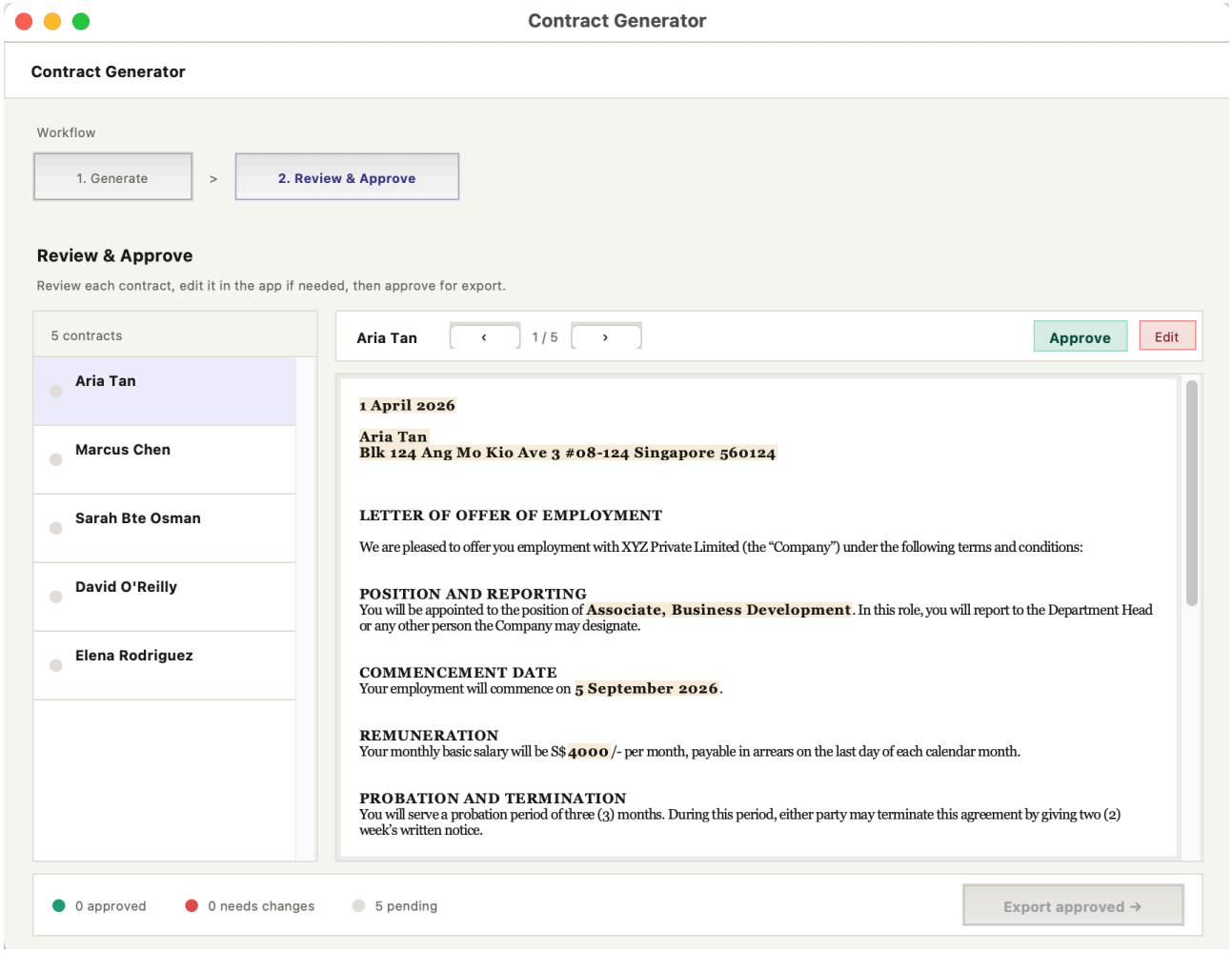
Click Browse to select where to output the Contract Generated Folder

Fourth Step (Generating Contracts)

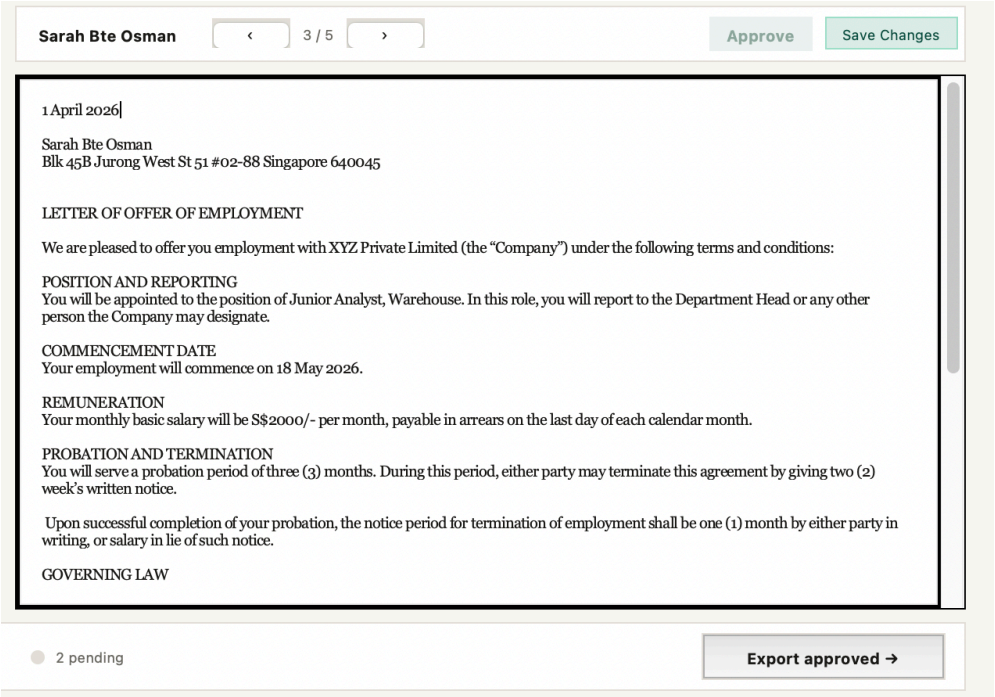


Click on Generate Contracts then Press Review & Approve

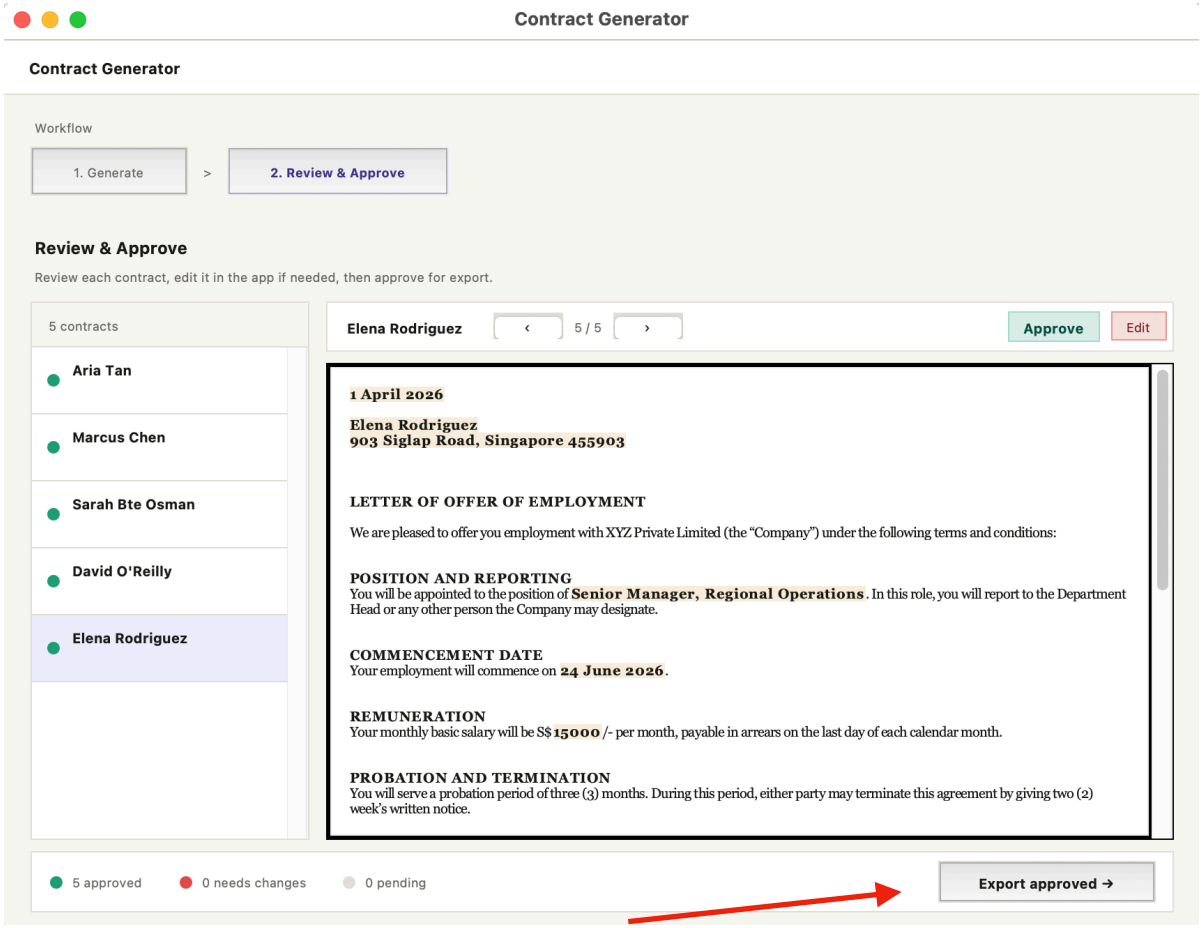
Fifth Step (Approve and Edit Generated Contracts)



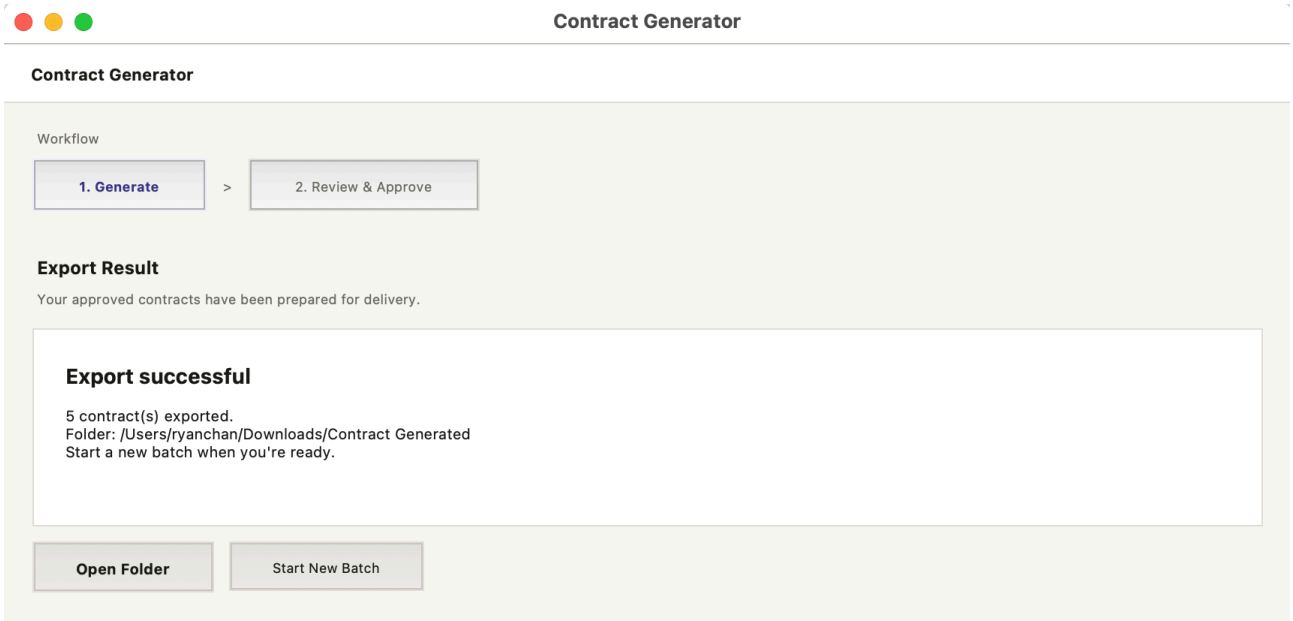
Final check for if fields have been correctly filled,
The user may edit the document if there are discrepancies, saving changes will refresh the preview. If there is no discrepancy, they can approve the document



Sixth Step (Export all approved contracts)

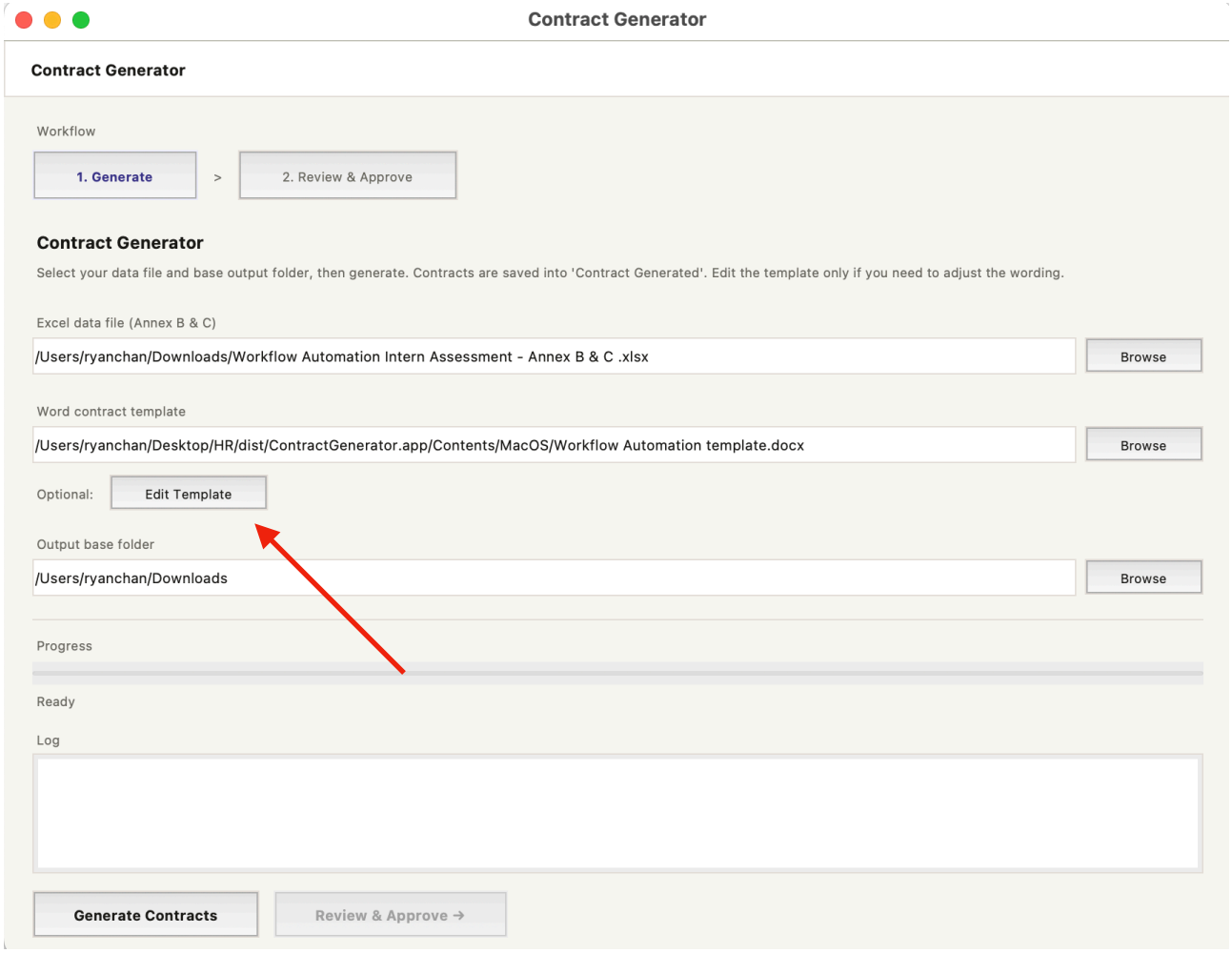


Click on Export Approved to export the approved documents
This may take a few seconds, there may be some terminal pop ups along the way as well
However, this is to be expected as the contracts are being converted to pdf via the backend



Click on Open Folder to view exported Contracts

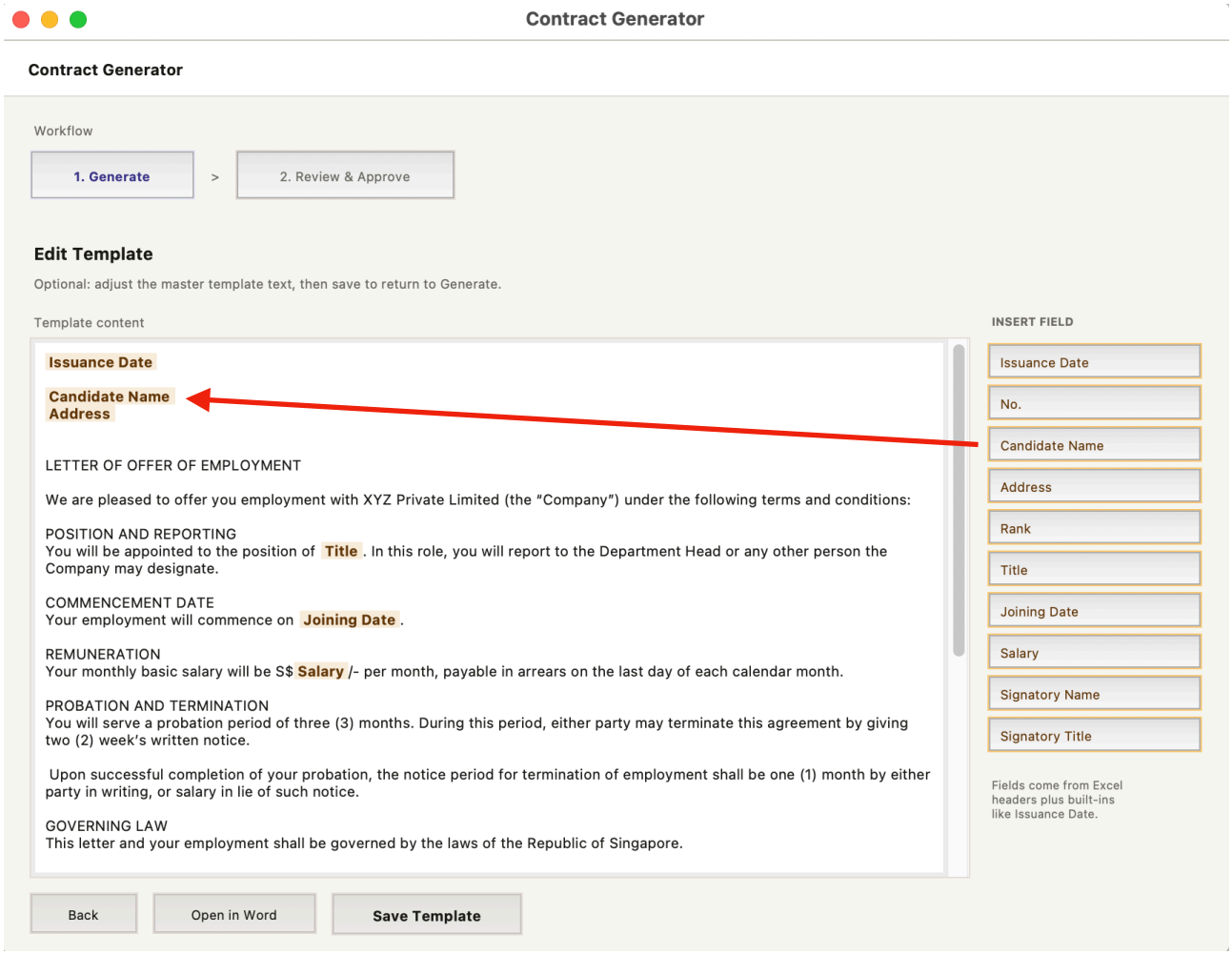
Editing Template Steps



Click Edit Template

Disclaimer: Import the xlsx file before editing template as autofilled fields are derived from the xlsx file

Edit template by inserting fields



The user can edit or revise the contract layout adding in further details and information should they choose to. The user can also add in the fields that are to be autofilled by clicking on the fields in the insert field column. Saving the template would permanently update the contract layout and return back to the generate page

That concludes the basic step by step guide on how to utilise this Contract Generator Application